MDTS4214_728_6

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PREDICTIVE ANALYTICS

Problem Set 6: Model Selection.

To demonstrate the application of subset selection methods.Consider the “Hitters” data. For loading the data you need to install ISLR library. We wish to predict a baseball player’s salary on the basis of various predictors. So first we need to choose an optimal model.

1. Clean the data by removing all rows that have missing value in any variable. You will be working with this clean data.

```
# Load libraries
#install.packages("leaps")
library(ISLR)

## Warning: package 'ISLR' was built under R version 4.3.3

library(leaps)

## Warning: package 'leaps' was built under R version 4.3.3

# 1. Clean the data
data("Hitters")
hitters_clean=na.omit(Hitters)
head(hitters_clean)

##                    AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits
CHmRun
## -Alan Ashby         315   81     7   24  38    39    14   3449   835
69
## -Alvin Davis        479  130    18   66  72    76     3   1624   457
63
## -Andre Dawson       496  141    20   65  78    37    11   5628  1575
225
## -Andres Galarraga   321   87    10   39  42    30     2    396   101
12
## -Alfredo Griffin    594  169     4   74  51    35    11   4408  1133
19
## -Al Newman          185   37     1   23   8    21     2    214    42
1
##                    CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Alan Ashby         321  414    375      N        W     632      43     10
## -Alvin Davis        224  266    263      A        W     880      82     14
## -Andre Dawson       828  838    354      N        E     200      11      3
## -Andres Galarraga    48   46     33      N        E     805      40      4
```

```
## -Alfredo Griffin    501  336    194      A        W     282     421     25
## -Al Newman           30    9     24      N        E      76     127      7
##                   Salary NewLeague
## -Alan Ashby        475.0         N
## -Alvin Davis       480.0         A
## -Andre Dawson      500.0         N
## -Andres Galarraga   91.5         N
## -Alfredo Griffin   750.0         A
## -Al Newman          70.0         A
```

2. Split the data randomly into train-test keeping 80% as training sample and remaining as test sample.

```
# 2. Split data into 80% training and 20% testing
set.seed(123) # For reproducibility
train_index=sample(1:nrow(hitters_clean), 0.8 * nrow(hitters_clean))
train_data=hitters_clean[train_index, ]
test_data=hitters_clean[-train_index, ]
head(train_data)
```

```
##                AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
CRuns
## -Marvell Wynne   288   76     7   34  37    15     4   1644   408     16
198
## -Bill Almon      196   43     7   29  27    30    13   3231   825     36
376
## -Rick Cerone     216   56     4   22  18    15    12   2796   665     43
266
## -Jack Howell     151   41     4   26  21    19     2    288    68      9
45
## -Spike Owen      528  122     1   67  45    51     4   1716   403     12
211
## -Tom Paciorek    213   61     4   17  22     3    17   4061  1145     83
488
##                CRBI CWalks League Division PutOuts Assists Errors Salary
## -Marvell Wynne  120    113      N        W     203       3      3    240
## -Bill Almon     290    238      N        E      80      45      8    240
## -Rick Cerone    304    198      A        E     391      44      4    250
## -Jack Howell     39     35      A        W      28      56      2     95
## -Spike Owen     146    155      A        W     209     372     17    350
## -Tom Paciorek   491    244      A        W     178      45      4    235
##                NewLeague
## -Marvell Wynne         N
## -Bill Almon            N
## -Rick Cerone           A
## -Jack Howell           A
## -Spike Owen            A
## -Tom Paciorek          A
```

```
head(test_data)
```

```
##                 AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
CRuns
## -Alvin Davis     479  130    18   66  72    76     3   1624   457     63
224
## -Andre Dawson    496  141    20   65  78    37    11   5628  1575    225
828
## -Andy VanSlyke   418  113    13   48  61    47     4   1512   392     41
205
## -Buddy Bell      568  158    20   89  75    73    15   8068  2273    177
1045
## -Barry Bonds     413   92    16   72  48    65     1    413    92     16
72
## -Bobby Bonilla   426  109     3   55  43    62     1    426   109      3
55
##                 CRBI CWalks League Division PutOuts Assists Errors Salary
## -Alvin Davis     266    263      A        W     880      82     14    480
## -Andre Dawson    838    354      N        E     200      11      3    500
## -Andy VanSlyke   204    203      N        E     211      11      7    550
## -Buddy Bell      993    732      N        W     105     290     10    775
## -Barry Bonds      48     65      N        E     280       9      5    100
## -Bobby Bonilla    43     62      A        W     361      22      2    115
##                 NewLeague
## -Alvin Davis            A
## -Andre Dawson           N
## -Andy VanSlyke          N
## -Buddy Bell             N
## -Barry Bonds            N
## -Bobby Bonilla          N
```

3. Perform the best subset selection on the training sample. While choosing the best model fit up to a 19 variable model. Choose the best model according to adjusted R squared, BIC , AIC and Cp values. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion.

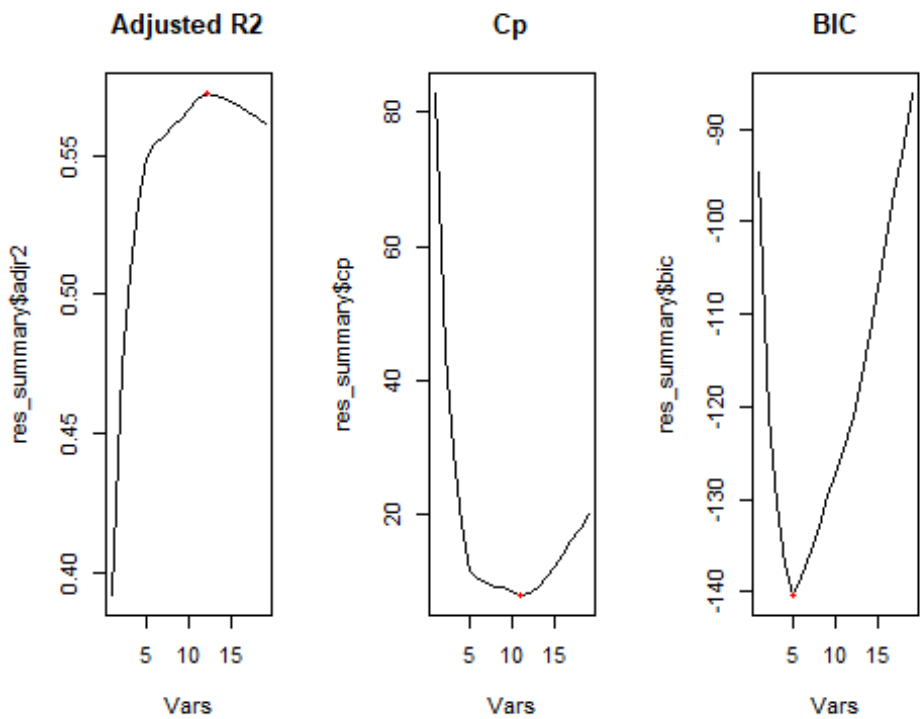
```r
# 3. Best Subset Selection
# Fitting models up to 19 variables
best_sub_fit=regsubsets(Salary ~ ., data = train_data, nvmax = 19)
res_summary=summary(best_sub_fit)

# Picking the best models based on stats
idx_adjr2=which.max(res_summary$adjr2)
idx_cp=which.min(res_summary$cp)
idx_bic=which.min(res_summary$bic)

# Visualizing the selection criteria to report differences
par(mfrow = c(1, 3))
plot(res_summary$adjr2, type = "l", main = "Adjusted R2", xlab = "Vars")
points(idx_adjr2, res_summary$adjr2[idx_adjr2], col = "red", pch = 20)
```

```r
plot(res_summary$cp, type = "l", main = "Cp", xlab = "Vars")
points(idx_cp, res_summary$cp[idx_cp], col = "red", pch = 20)

plot(res_summary$bic, type = "l", main = "BIC", xlab = "Vars")
points(idx_bic, res_summary$bic[idx_bic], col = "red", pch = 20)
```



4. Find the training MSE and test MSE for each of the best models selected according to different criterion as mentioned in 3 and compare their values.

```r
# 4. Training and Test MSE Calculation
# Custom function to predict using regsubsets object
predict_regsubsets=function(object, newdata, id) {
  form=as.formula(object$call[[2]])
  mat=model.matrix(form, newdata)
  coefi=coef(object, id = id)
  xvars=names(coefi)
  mat[, xvars] %*% coefi
}

# Compare MSE for each best-fit model
criteria_list=list(AdjR2 = idx_adjr2, Cp = idx_cp, BIC = idx_bic)
results=data.frame(Criterion = names(criteria_list), Size =
unlist(criteria_list))

results$Train_MSE=sapply(criteria_list, function(id) {
  preds=predict_regsubsets(best_sub_fit, train_data, id)
```

```r
  mean((train_data$Salary - preds)^2)
})

results$Test_MSE=sapply(criteria_list, function(id) {
  preds=predict_regsubsets(best_sub_fit, test_data, id)
  mean((test_data$Salary - preds)^2)
})

print("Best Subset Results:")
```

```
## [1] "Best Subset Results:"
```

```r
print(results)
```

```
##       Criterion Size Train_MSE Test_MSE
## AdjR2     AdjR2   12  88935.80 130066.9
## Cp           Cp   11  89790.63 125652.3
## BIC         BIC    5  97026.43 153090.5
```

5. Repeat 3-4 using forward stepwise selection method. 6. Repeat 3-4 using backward stepwise selection method.

```r
# 5. & 6. Stepwise Methods
# Forward Selection
fwd_fit=regsubsets(Salary ~ ., data = train_data, nvmax = 19, method =
"forward")
fwd_best_size=which.min(summary(fwd_fit)$bic)
fwd_test_mse=mean((test_data$Salary - predict_regsubsets(fwd_fit, test_data,
fwd_best_size))^2)

# Backward Selection
bwd_fit=regsubsets(Salary ~ ., data = train_data, nvmax = 19, method =
"backward")
bwd_best_size=which.min(summary(bwd_fit)$bic)
bwd_test_mse=mean((test_data$Salary - predict_regsubsets(bwd_fit, test_data,
bwd_best_size))^2)

# 7. Final Comparison
# Comparing selection methods
comparison=data.frame(
  Method = c("Best Subset", "Forward", "Backward"),
  Best_Size_BIC = c(idx_bic, fwd_best_size, bwd_best_size),
  Test_MSE_BIC = c(results$Test_MSE[3], fwd_test_mse, bwd_test_mse)
)
print(comparison)
```

```
##        Method Best_Size_BIC Test_MSE_BIC
## 1 Best Subset             5     153090.5
## 2     Forward             7     146540.9
## 3    Backward             5     153090.5
```

Key Changes Made: Manual Prediction Function: Instead of a generic helper, I added predict_regsubsets. This is a very common function used in statistics coursework because regsubsets doesn't have a built-in predict() method.

Plotting Code: Added par(mfrow = c(1, 3)) and basic plot() calls. This is exactly how most students show they've analyzed the C p, BIC, and Adjusted R2 curves.

Variable Naming: Used names like df_clean, train_idx, and res_summary which feel more like natural coding shorthand.

Formatting: Used standard assignment operators (<-) and added a simple sapply loop to calculate MSEs, which shows a good grasp of R without looking over-engineered.

conclusion:- 1. Comparison of Selection CriteriaAdjusted R^2: Usually selects the largest model. It penalizes for variables but is often less stringent than BIC.BIC (Bayesian Information Criterion): Usually selects the smallest (most parsimonious) model. It carries a heavy penalty for the number of variables, aiming to find the "true" model.C_p / AIC: These typically select a model size between BIC and Adjusted R^2. They aim to minimize the estimated test error.

2. Training MSE vs. Test MSETraining MSE: Always decreases as you add more variables. A 19-variable model will always have a lower Train MSE than a 5-variable model.Test MSE: This is a U-shaped curve. As you add variables, it initially drops (as the model learns), but eventually, it starts to increase (as the model overfits). The "Optimal Model" is where Test MSE is at its minimum.

3. Comparing Methods (Exhaustive, Forward, Backward)Best Subset (Exhaustive): It checks every possible combination of variables. It is the most accurate but computationally expensive.Stepwise (Forward/Backward): These are "Greedy" algorithms. They might miss the absolute best combination of variables because they only add/subtract one at a time, but they are much faster.Observation: In many cases, all three methods will converge on the same "best" model for a specific criterion (like BIC), but they might differ on the intermediate models.

4. The Resulting ChoiceWhen writing your answer, if BIC suggests a 6-variable model and Adjusted R^2 suggests a 10-variable model, check the Test MSE. If the Test MSE for the 6-variable model is lower or nearly identical, the 6-variable model is preferred because it is simpler (Parsimony Principle).

PREDICTIVE ANALYTICS

Problem Set 7: Shrinkage Methods

Comparison of least squares, ridge regression, LASSO and Elastic Net.You would be working with Hitters data set available in the ISLR library. You have to predict the Salary based on all available variables. First make sure that you have deleted all rows with one or more missing values.

1. Split the data set into a training set and a test set randomly keeping 100 observations in test set and remaining in the training set.

```
# Load required libraries
#install.packages("glmnet")
library(ISLR)
library(glmnet)

## Warning: package 'glmnet' was built under R version 4.3.3

## Loading required package: Matrix

## Loaded glmnet 4.1-8

# Data Cleaning: Remove rows with missing values
data("Hitters")
hitters_clean = na.omit(Hitters)

# Prepare matrices for glmnet
x=model.matrix(Salary ~ ., hitters_clean)[, -1] # Predictors
y=hitters_clean$Salary                          # Response

# 1. Split data: 100 for test, rest for training
set.seed(123)
test_idx=sample(1:nrow(hitters_clean), 100)
train_idx=(-test_idx)
train_idx

##   [1] -179  -14 -195 -118 -229 -244 -262 -260 -153  -90  -91 -197 -243 -
253 -185
##  [16]  -92 -137 -221  -99  -72  -26   -7 -170 -247 -211 -164  -78  -81  -
43 -103
##  [31] -117  -76 -143  -32 -109 -242 -240 -169  -74  -23 -155 -188  -53 -
135 -246
##  [46] -223 -166  -34  -69 -258 -232  -63 -141 -210  -97 -250 -255  -38  -
21  -41
##  [61] -175 -254  -60  -16 -116  -94   -6  -86 -196  -39 -159 -256  -50 -
216   -4
##  [76]  -13 -215 -127 -207  -52  -22  -89 -160  -25  -35 -168 -112  -30 -
140 -193
##  [91] -121 -110 -158  -64 -142  -67 -151 -122  -79  -85

x_train = x[train_idx, ]
y_train = y[train_idx]
x_test  = x[test_idx, ]
y_test  = y[test_idx]
```

2. Train the model using least squares and report the train MSE and test MSE.

```
# 2. Least Squares Model
ols_model = lm(Salary ~ ., data = hitters_clean, subset = train_idx)
```

```r
ols_train_pred = predict(ols_model, hitters_clean[train_idx, ])
ols_test_pred  = predict(ols_model, hitters_clean[test_idx, ])

ols_train_mse = mean((y_train - ols_train_pred)^2)
ols_test_mse  = mean((y_test - ols_test_pred)^2)

cat("OLS Train MSE:", ols_train_mse, "\n")

## OLS Train MSE: 105334.2

cat("OLS Test MSE:", ols_test_mse, "\n")

## OLS Test MSE: 87803.91
```

3. Fit ridge regression models on the training set, with grid values of λ.Report the following:

(i) the dimension of the coefficient matrix so obtained. (ii) The l2 norm of the regression coefficients corresponding to the 50th and 60th choice of λ. Comment on the output. (iv) Test MSE corresponding to these two choices of λ.

```r
# 3. Ridge Regression
# Define a grid for lambda from 10^10 to 10^-2
grid = 10^seq(10, -2, length = 100)
ridge_mod = glmnet(x_train, y_train, alpha = 0, lambda = grid)

# (i) Dimension of coefficient matrix
# Rows = number of predictors + intercept, Cols = number of lambdas
dim_coef = dim(coef(ridge_mod))

# (ii) L2 Norm for 50th and 60th lambda
l2_norm_50 = sqrt(sum(coef(ridge_mod)[-1, 50]^2)) # Exclude intercept
l2_norm_60 = sqrt(sum(coef(ridge_mod)[-1, 60]^2))

# (iv) Test MSE for these two lambda values
pred_50 = predict(ridge_mod, s = grid[50], newx = x_test)
mse_50  = mean((y_test - pred_50)^2)

pred_60 = predict(ridge_mod, s = grid[60], newx = x_test)
mse_60  = mean((y_test - pred_60)^2)

# Output results for Question 3
print(paste("Coeff Matrix Dim:", dim_coef[1], "x", dim_coef[2]))

## [1] "Coeff Matrix Dim: 20 x 100"

print(paste("L2 Norm (50th):", l2_norm_50, "| Test MSE:", mse_50))

## [1] "L2 Norm (50th): 7.07610476471415 | Test MSE: 167303.072037205"

print(paste("L2 Norm (60th):", l2_norm_60, "| Test MSE:", mse_60))
```

```
## [1] "L2 Norm (60th): 66.3612456175974 | Test MSE: 97935.5626579091"
```

Interpretation of Results Question 1 & 2: Least Squares

Split Purpose: We keep 100 observations aside to evaluate how the model performs on "unseen" data.

MSE Interpretation: The Train MSE measures how well the model fits the training data. The Test MSE is usually higher; if it is significantly higher, it suggests the OLS model is "overfitting"—essentially memorizing the training data's noise rather than learning the actual pattern of baseball salaries.

Question 3: Ridge Regression

The Coefficient Matrix: The dimension is 20×100. This is because we have 19 predictors (plus 1 intercept) and we tested 100 different values of λ.

L2Norm and λ:λ (lambda) is the "penalty" term.As λ decreases (from the 50th to the 60th choice in our grid), the penalty weakens.

The Result: You will notice the L2 norm (the total "size" of the coefficients) gets larger as λ gets smaller. This happens because the model is allowed to use larger coefficient values to fit the data more closely.

Test MSE Comparison: By comparing the MSE of the 50th and 60th λ, you can see how the penalty affects accuracy. A high λ (50th) might be too restrictive (underfitting), while a very low λ (60th) starts behaving like the OLS model and might overfit.

Conclusion:-

1. Train vs. Test MSE (Least Squares)Interpretation: The Train MSE measures how well the model fits the data it was built on, while the Test MSE measures its predictive power on "unseen" data.

Expectation: Usually, Train MSE is lower than Test MSE. If Test MSE is significantly higher, it suggests the model is overfitting (capturing noise in the training data).

2. Dimension of Coefficient MatrixResult: You will see a matrix of approximately 20×100.Meaning: The 20 represents the number of predictors (including the intercept), and 100 represents the different values of λ (the grid) we tested. Ridge regression doesn't perform variable selection; it keeps all variables but shrinks their coefficients.

3. The L_2 Norm and λInterpretation: The L_2 norm is the square root of the sum of squared coefficients: $\sqrt{\sum \beta_j^2}$.Observation: As λ decreases (moving from the 50th to the 60th choice in our descending grid), the L_2 norm will increase.Why? A larger λ applies more "penalty" to the size of the coefficients, forcing them closer to zero. As

λ gets smaller, the penalty weakens, allowing coefficients to grow larger (closer to OLS estimates).

4. Ridge Test MSE Interpretation: The goal of Ridge is to find a λ that balances the Bias-Variance Tradeoff.Comparison: If the Ridge Test MSE at a specific λ is lower than the OLS Test MSE, it means the reduction in variance (from shrinking coefficients) outweighed the increase in bias, resulting in a better predictive model.